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Laboratory work 2:
Cryptanalysis of Monoalphabetic Ciphers
(Frequency Analysis Attack, Variant 16)

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1 ALGORITHM ANALYSIS

1.1 Objective

Study the frequency analysis attack on monoalphabetic substitution ciphers and apply it to recover the plaintext and the key of an intercepted English cryptogram (variant 16).

1.2 Tasks

An encrypted message was intercepted. It is known that it was obtained with a monoalphabetic cipher and that the plaintext is written in English. Only the letters were encrypted; spaces, digits and punctuation were left unchanged. The task is to find the original message by a frequency analysis attack and to describe the breaking process step by step, in the same way as in section 2.3 of the laboratory guide. In addition, the key used for encryption has to be recovered.

1.3 Theoretical Notes

A **monoalphabetic substitution cipher** replaces every plaintext letter with one and the same ciphertext letter throughout the whole message. The key is a permutation of the alphabet, so for English there are $26! \approx 4 \cdot 10^{26}$ keys and an exhaustive search is impossible. The weak point of these ciphers is that they keep the **frequency of the letters**: if E is the most common letter in English and it is encrypted as V, then V will be the most common letter of the ciphertext. The longer the ciphertext, the closer its letter frequencies are to the general frequencies of the language (Table 1).

Table 1: Frequency of English letters, % (Table 2.2 of the laboratory guide).

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
%	8.17	1.49	2.78	4.25	12.70	2.23	2.01	6.09	6.97	0.15	0.77	4.03	2.41
Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
%	6.75	7.51	1.93	0.09	5.99	6.33	9.06	2.76	0.98	2.36	0.15	1.97	0.07

Frequencies alone rarely give the whole key, because several letters have close frequencies (for example T, A and O). The analyst also uses other “personality traits” of the language: the most common words (*the, and, of, to, a, in*), the only one-letter words *a* and *I*, frequent digraphs (TH, HE, AN, IN, ER) and trigraphs (THE, AND, ING), and partly decrypted words whose missing letters can be guessed from the context. For this reason the attack is an interactive process: the computer counts and substitutes, and the human makes the guesses.

2 Introduction

Frequency analysis is the oldest known cryptanalytic technique. Its first description is found in the manuscript of Al-Kindi, written around 850 AD. Even today it explains why every cipher that maps one plaintext symbol to one fixed ciphertext symbol is insecure, no matter how large its key space is.

2.1 Intercepted Cryptogram

The ciphertext of variant 16 contains 3239 letters:

Wqv cxiw ngu rtp rtxwgg xg wqv rggjp. Wqxp rtp unsf-tsuqtavwxhpdapwxdwng, xg wqv cniz nc wqv pwtixjqw-tsuqtavw Kxjvgviv rrwq pqniwivwtwgg lufnio. Wqv nso naevhwngp wn xwp dpv, rqxhq anxsv onrg wnwqv xzunppxaxswf nc hniivhwgg t jtiasvo oxputwhq bdxhlsf vgnjdj,ktgxpqvo rrwq wqv wsvjituq. Xw cdscxssvo wqv ivbdxivzugw nc gnghnz-uinzxptaxswf nc wqv jugvits pfpwvz tgo nc vtpv nc luf hqtgjp. Znivnkvi,xw qto wqv ivudwtwng nc avxgj dgaivlttasv—rqxhq, xc xwp hifuwnjitzprviv gnw oxkxovo xgun rniop, xw stijvsf rtp. Wqv xaxwtif tonuwvo xw twnghv.Wqvg, xg 1863, t ivwxivo Uidppxtg xgctgwif zteni oxphnkvivo wqvjugvits pnsdwng cni wqv uvixnozh unsftsuqt-avwxh pdapwxdwng. Tw ngvpinlv qv ovznxqpvo wqv ngvf xzuivjtasv pwidhwdv xg hifuwnjituqf.Pxjgts nccxhvip, hnzuvsso wn uinkxov puhdiv hnzzdgxhtwngp, qdgvvocitgwxtssf cni gvr cxso hruqip. Wqv cndgo ztgf jnno xoutp xg wqvrixwggp nc wqv oxsvwwtgw hifuwnjituqvip rqn qto uinunpvo hruqip cniwqv uinwvhwng nc uixktwv zvpptjp. Pnng pnzv nc wqvvp pfpwvzp rviupvikxg xg wqv ktixndp tixvp nc Vdinuv tgo wqv Tzixhtp. Zniv xoutphtzv cinz tizf nccxhvip rqn qto pwxovo hifuwnjituqf xg wqv hndipvp xgpjgts hnzzdgxhtwng wgtw wqv gtwngts xaxwtif thtovxvp, pdhq tp Pu.Hfi, qto toovo xg wqv xzo-1800p. Xgukwtasf, hifuwtgtsfwp—rqn rviuvxwvi tztwdip ni pnsorvip rrwq t wincvppxngts xgwviwp, cni cdssuincvppxngtsp wqviv rivv gngv—ivusxo rrwq gvr wvhqgxbdv cni aivtlxgjqv gvr hruqip. Cinz wqv psnr hitrs nc gnvghstwni oftp, rvg wqv xgwinodhwng nc t puvhats jindu ztgxgj Oxpijivio wqv uivhvo xg jindurndso hngpwxwdu t ivziltasv wvhqgchts tokthv, wqv ithv avwrvgnccvvpv tgo ovcvgpv xg hifuwnsnjf thhsvitwvo wn xwp znovig uthv.Wqv qxpunif nc hifuwnsnjf cinz wqv ovhtov wgtw ptr anwq wqv outwq ncwqv asthl hqtzavip tgo wqv axiwq nc wqv wsvjituq wn Rniso Rti X xp wqdp pwnif nc xgwigts ovkusnuzvgw. Rrwqndw Inppxjgns ni Rssvvpv, tgorxwqndv zteni rtip ni oxusntwv pwidjisp, hifuwnsnjf hndso gnwxgcsdvghv rniso vkvgw, tgo, vyhvuv cni ngu ni wrn dgdpts htpv, xw oxognw. Wqv wsvjituq stdghqvo wqx vknsdwng nc hifuwnsnjf. Xw ainlv wqv zngnunsf nc wqv gnvghstwni. Wqvgnvghstwni qto ivxjvo cni 450 futip tp t jugvits, tss-udiunpv pfpwvz,adw xw hndso gnw zvvw wqv gvr ivbdxivzugw vxwvi nc qxjq-sukvsorxusntwv ni xaxwtif hnzzdgxhtwngp ni nc snr-sukvs pxjgts hnzzdgxhtwngp, rqxhq wqv wsvjituq qto vgjgovivo. Vthq htssvo cni xwpnrg lrgo nc hifuwnpfpwvz, t puvhatsxmvo ngu. Pxjgts nccxhvip itglvo wqvvpfpwvzp xg t qxivtihqf, ixp xg cinz wqv paxusv tgo csvyxasv tgo vtpxsfpnskvo wn wqv vywvpxkv tgo qtio wn pnskv. Wqv wsvjituq wqdp pwxzdstvowqv xgkvwng nc ztgf gvr hruqip tgo, af ivthwng, ztgf gvr zvwqnopnc hifu- tgtsfpx, tgo hnzuvsso wqvxi tiitgjvzvgw xg t phtsv nchnzusvyxw.Ztgf nc wqvvp hruqip tgo wvhqgxbdv qtkv avhnzv hstppv tgo tiv xgdpv wnotf. Znivnkvi, hifuwnjituqf pwxss cdghwngp wqindj t qxivtihqftgo vzusnf t zdsxwdov nc puvhats pfpwvzp. Wqv wsvjituq wqvivafcdigxqpvo hifuwnjituqf rrwq wqv pwidhwdv tgo wqv hngwvgw wgtw xw pwxssqtp. Xw ztov xw rqtw xw xp wnotf. Tss wqvvp wqxgjp qtkv tgvhvovgwp, tgo edpw tp wqv wsvjituq xwpvsc oxo,pn rivv wqv uivhdipniv nc wqv hifuwnjituqzh pfpwvzp wgtw xwvgjgovivo.Ngv hruqip pfpwvz xgkvwvo avcniv wqv wsvjituq rtp pn cti tqto ncrwp wxzv, tgo pn zdhq xg wqv puxixw nc wqv stwvi xgkvwngp, wgtw xw ovpvikvpwn av hstppvo rrwq wqvz. Xgovvo, xw ovpvikvp wqv cingw itgl tznj wqvz,cni wqx pfpwvz rtp avfngo ondaw wqv znpw ivziltasv nc tss. Pn russhnghvzkvo rtp xw wgtw wnotf, zniv wgtg t hvwdif tgo t qtsc nc ituxovhggnsnjxhts uinjivvp tcwvi xwp xgkvwng, xw ivtxgp xg thwkv dpv.Adw wqvg xw rtp xgkvwvo af t ivziltasv ztg, t russ-lgnrg rixwvi,tjixhdswditsxpw,

*axasxnuqxsu, tihqrxvhw, oxusnztw, jtojvvvvi, tgo pwtwvpztggtzvo Wqnztp
Evccvipng. Qv htssvo xw qxp "rqvvs hfuvvi," tgo xw pvvzpsxlvsf wqtw qv xgkvguwo
xw vxwqvi odixgj 1790 wn 1793 ni odixgj 1797 wn1800.*

2.2 Tools

The laboratory guide recommends the online service crypto.interactive-maths.com. To make every step reproducible, I wrote a small Python program, `frequency_analysis.py`, that does the same work: it counts the letters, compares them with the English frequencies, applies the guessed substitutions and shows the partly decrypted text after every guess. As in the guide, the letters that are still encrypted are shown in UPPER case and the guessed plaintext letters in lower case.

3 IMPLEMENTATION

3.1 The Analysis Tool

```
1 def letter_counts(text):
2     """Number of occurrences of every letter A-Z (case-insensitive)."""
3     counts = Counter(ch for ch in text.upper() if ch in ALPHABET)
4     return {letter: counts.get(letter, 0) for letter in ALPHABET}
5
6 def sorted_by_frequency(counts):
7     return sorted(counts, key=lambda letter: (-counts[letter], letter))
```

Listing 1: Counting the letters of the ciphertext

`letter_counts` ignores spaces, digits and punctuation, which were not encrypted, and counts every letter independently of its case. `sorted_by_frequency` gives the letters from the most to the least frequent, the order used in Table 3.

```
1 def partial_decrypt(text, mapping):
2     result = []
3     for ch in text.upper():
4         result.append(mapping.get(ch, ch) if ch in ALPHABET else ch)
5     return "".join(result)
6
7 def add_substitution(mapping, cipher, plain):
8     cipher, plain = cipher.upper(), plain.lower()
9     for other_cipher, other_plain in mapping.items():
10        if other_plain == plain and other_cipher != cipher:
11            return f"'{plain}' is already the image of {other_cipher}"
12    mapping[cipher] = plain
13    return None
14
15 def recover_key(mapping):
16     inverse = {plain: cipher for cipher, plain in mapping.items()}
17     return {plain: inverse.get(plain, "?") for plain in ALPHABET.lower()}
```

Listing 2: Applying the guesses and recovering the key

`mapping` stores the guesses (ciphertext letter $\rightarrow$ plaintext letter). `partial_decrypt` replaces only the guessed letters, so the analyst sees which letters are still unknown. `add_substitution` rejects a guess that would map two different ciphertext letters to the

same plaintext letter, since a substitution key is a permutation. `recover_key` inverts the mapping to obtain the key in the form of Table 2.5 of the guide. The program can be used interactively (commands such as `V=e W=t`) or can replay all the steps from the file `variant16_steps.txt`.

3.2 Letter Frequencies of the Cryptogram

The first step is to count all the letters of the cryptogram (Table 2) and to order them by decreasing frequency (Table 3).

Table 2: Frequency of the letters in the intercepted cryptogram.

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Count	38	5	78	72	5	75	203	117	214	67	32	16	1
Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Count	237	130	206	177	58	142	235	83	405	310	239	4	90

Table 3: Letters of the cryptogram in decreasing order of frequency, compared with English.

Ciphertext	V	W	X	N	T	I	P	G	Q	S	O	H	Z
Count	405	310	239	237	235	214	206	203	177	142	130	117	90
%	12.5	9.6	7.4	7.3	7.3	6.6	6.4	6.3	5.5	4.4	4.0	3.6	2.8
English	E	T	A	O	I	N	S	H	R	D	L	C	U
Ciphertext	U	C	F	D	J	R	A	K	L	B	E	Y	M
Count	83	78	75	72	67	58	38	32	16	5	5	4	1
%	2.6	2.4	2.3	2.2	2.1	1.8	1.2	1.0	0.5	0.2	0.2	0.1	0.0
English	M	W	F	G	Y	P	B	V	K	J	X	Q	Z

Figures 1 and 2 show the two distributions. The shapes are clearly similar, only shifted to other letters, which confirms that a monoalphabetic cipher was used.

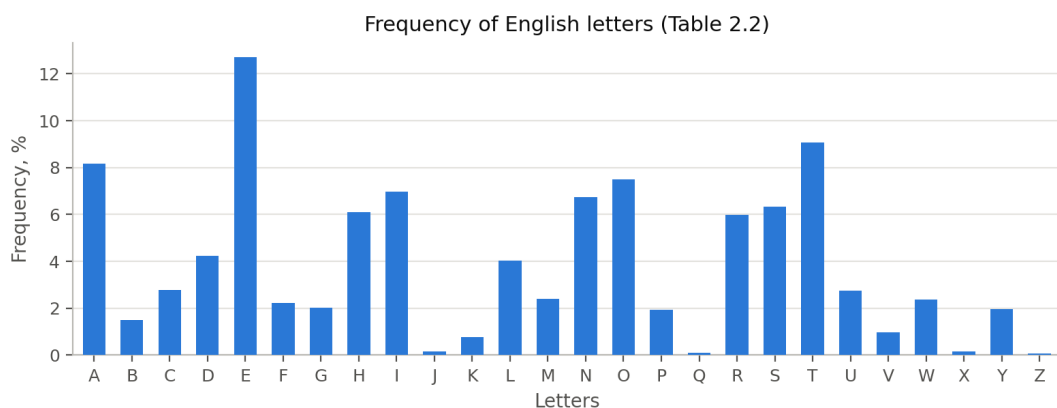


Figure 1: Frequency of the letters of the English language.

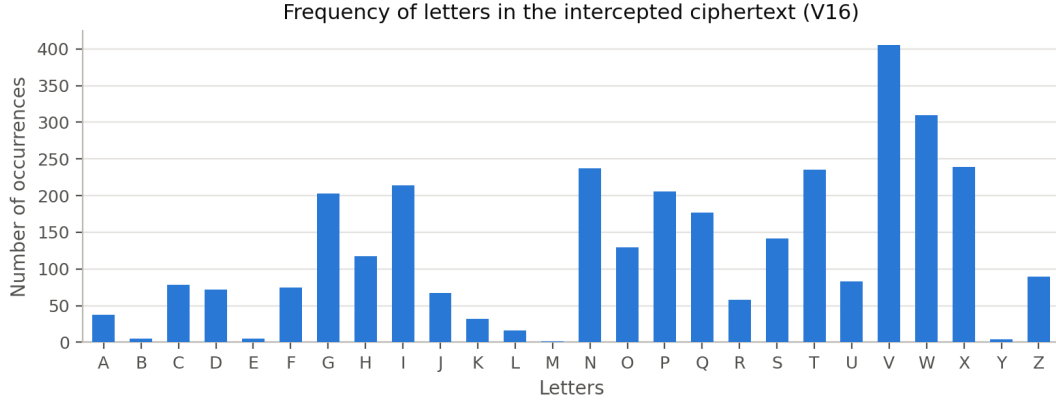


Figure 2: Frequency of the letters in the intercepted cryptogram.

3.3 Breaking the Cipher Step by Step

To keep the report readable, after every step only the first five sentences of the cryptogram are shown; the program applies each substitution to the whole text. Lower-case letters are already decrypted, upper-case letters are still ciphertext. The words that suggested the next substitution are shown in bold.

Step 1. The most frequent letter of the cryptogram is V (405 times, 12.5%), followed by W (310 times, 9.6%). In English the two most frequent letters are E (12.70%) and T (9.06%), so we assume $V \rightarrow e$ and $W \rightarrow t$:

*tQe CXIPt NGe RTP RTXtXGJ XG tQe RXGJP. tQXP RTP
UNSF-TSUQTAetXHPDAPtXtDtXNG, XG tQe CNIZ NC tQe
PtITXJQt-TSUQTAet KXJeGeIe RXtQ PQNItIeUeTtXGJ LeFRNIO. tQe NSO
NAEeHtXNGP tN XtP DPe, RQXHq ANXSeO ONRG tNtQe
XZUNPPXAXSxtF NC HNIIeHtXGJ T JTIASeO OXPUTtHQ BDXHLSF
eGNDJQ,KTGXpQeO RXtQ tQe teSeJITUQ. Xt CDSCXSSeO tQe
IeBDXIeZeGtP NC GNGHNZ-UINZXPTAXSxtF NC tQe JeGeITS PFPteZ
TGO NC eTPe NC LeF HQTGJeP. ZNIeNKeI,Xt QTO tQe IeUDtTtXNG NC
AeXGJ DGAIEtLTASe—RQXHq, XC XtP HIFUtNJITZPreIe GNt OXKXOeO
XGtN RNIOP, Xt STIJeSF RTP.*

Step 2. The word “tQe” appears 44 times, more often than any other word. The most common three-letter word in English is “the”, which agrees with the substitutions already made, so $Q \rightarrow h$:

*the CXIPt NGe RTP RTXtXGJ XG the RXGJP. thXP RTP
UNSF-TSUhTAetXHPDAPtXtDtXNG, XG the CNIZ NC the
PtITXJht-TSUhTAet KXJeGeIe RXth PhNItIeUeTtXGJ LeFRNIO. the NSO
NAEeHtXNGP tN XtP DPe, RhXHh ANXSeO ONRG tNthe XZUNPPXAXSxtF
NC HNIIeHtXGJ **T** JTIASeO OXPUTtHh BDXHLSF eGNDJh,KTGXPhO
RXth the teSeJITUh. Xt CDSCXSSeO the IeBDXIeZeGtP NC
GNGHNZ-UINZXPTAXSxtF NC the JeGeITS PFPteZ **TGO** NC eTPe NC LeF
HhTGJeP. ZNIeNKeI,Xt hTO the IeUDtTtXNG NC AeXGJ
DGAIEtLTASe—RhXHh, XC XtP HIFUtNJITZPreIe GNt OXKXOeO XGtN
RNIOP, Xt STIJeSF RTP.*

Step 3. The letter T stands alone as a word 15 times. The only one-letter English words are “a” and “I”, and T is the fifth most frequent letter, so $T \rightarrow a$. The second most frequent word is “TGO” (18 times), which with $T = a$ becomes “aGO” and is clearly “and”. Therefore $G \rightarrow n$ and $O \rightarrow d$:

*the CXIPt Nne RaP RaXtXnJ **Xn** the RXnJP. thXP RaP*
*UNSF-aSUhaAetXHPDAPtXtDtXNn, Xn the CNIZ **NC** the PtIaXJht-aSUhaAet*
KXJeneIe RXth PhNIeIeUeatXnJ LeFRNId. the NSd NAEeHtXNnP tN XtP DPe,
RhXHh ANXSed dNRn tNthe XZUNPPXAXSXtF NC HNIIeHtXnJ a JaIASed
*dXPUatHh BDXHLSF enNDJh,KanXPhed RXth the teSeJIaUh. **Xt** CDSCXSsed*
the IeBDXIeZentP NC nNnHNZ-UINZXPaAXSXtF NC the JeneIaS PFPteZ and
NC eaPe NC LeF HhanJeP. ZNIeNKeI,Xt had the IeUDtatXNn NC AeXnJ
DnAleaLaASe—RhXHh, XC XtP HIFUtNJIaZPreIe nNt dXKXded XntN
RNIIdP, Xt SaIJeSF RaP.

Step 4. The two-letter word “NC” appears 28 times. The most common two-letter English word is “of”, so $N \rightarrow o$ and $C \rightarrow f$. The words “Xt” (19 times) and “Xn” (13 times) can only be “it” and “in”, so $X \rightarrow i$. The frequencies agree: N, X and G are among the most frequent ciphertext letters, just like o, i and n in English:

the fIPt one RaP RaitinJ in the RinJP. thiP RaP
UoSF-aSUhaAetiHPDAPtitDtion, in the foIZ of the PtIaiJht-aSUhaAet KiJeneIe
Rith PhoItIeUeatinJ LeFRoId. the oSd oAEeHtionP to itP DPe, RhiHh AoiSed
doRn tothe iZUoPPiAiSitF of HoIIeHtinJ a JaIASed diPUatHh BDiHLSF
*enoDjh,KaniPhed Rith the **teSeJIaUh**. it fDSfiSSed the IeBDiLeZentP of*
nonHoZ-UIoZiPaAiSitF of the JeneIaS PFPteZ and of eaPe of LeF HhanJeP.
ZoIeoKeI,it had the IeUDtation of AeinJ DnAleaLaASe—RhiHh, if itP
HIFUtoJIaZPreIe not diKided into RoIdP, it SaIJeSF RaP.

Step 5. The word “teSeJIaUh” appears 8 times. The only English word that fits this pattern is “telegraph”, which also fits the subject of the text. This gives four substitutions at once: $S \rightarrow l$, $J \rightarrow g$, $I \rightarrow r$ and $U \rightarrow p$:

*the firPt one **RaP Raiting** in the **RingP**. thiP RaP*
polF-alphaAetiHPDAPtitDtion, in the forZ of the Ptraight-alphaAet Kigenere
***Rith** Phortrepeating LeFRord. the old oAEeHtionP to itP DPe, RhiHh Aoiled*
doRn tothe iZpoPPiAilitF of HorreHting a garAled diPpatHh BDiHLIF
enoDgh,KaniPhed Rith the telegraph. it fDIfilled the reBDireZentP of
nonHoZ-proZiPaAilitF of the general PFPteZ and of eaPe of LeF HhangeP.
ZoreoKer,it had the repDtation of Aeing DnAreaLaAle—RhiHh, if itP
HrFptograZPRere not diKided into RordP, it largelF RaP.

Step 6. The text now contains “Rith” (6 times) and “Raiting in the RingP”, which suggest “with” and “waiting in the wings”, so $R \rightarrow w$. Then “waP” (7 times) is “was” and “thiP” is “this”, so $P \rightarrow s$:

the first one was waiting in the wings. this was polF-alphaAetiHsDAstitDtion, in
*the **forZ** of the straight-alphaAet Kigenere with shortrepeating LeFword. the old*
oAEeHtions to its Dse, whiHh Aoiled down tothe iZpossiAilitF of HorreHting a
garAled dispatHh BDiHLIF enoDgh,Kanished with the telegraph. it fDIfilled the

*reBDireZents of nonHoZ-proZisaAilitF of the general **sFsteZ** and of ease of LeF Hhanges. ZoreoKer,it had the repDtation of Aeing DnAreaLaAle—which, if its HrFptograZswere not diKided into words, it largelF was.*

Step 7. The word “HrFptographF” (4 times) is “cryptography” and “HrFptologF” is “cryptology”, so $H \rightarrow c$ and $F \rightarrow y$. Then “systeZ” and “forZ” are “system” and “form”, so $Z \rightarrow m$:

*the first one was waiting in the wings. this was poly-alphaAeticsDAstitDtion, in the form of the straight-alphaAet **Kigenere** with shortrepeating Leyword. the old oAEctions to its **Dse**, which Aoiled down tothe impossiAility of correcting a **garAled** dispatch BDicLly enoDgh,**Kanished** with the telegraph. it fDIfilled the reBDirements of noncom-promisaAility of the general system and of ease of Ley changes. moreoKer,it had the repDtation of Aeing DnAreaLaAle—which, if its cryptogramswere not diKided into words, it largely was.*

Step 8. Only rare letters are left. “Dse” is “use” and “fDIfilled” is “fulfilled”, so $D \rightarrow u$. “Aeing” and “garAled” give “being” and “garbled”, so $A \rightarrow b$. “Kigenere”, “Kanished” and “deKelopment” are “Vigenere”, “vanished” and “development”, so $K \rightarrow v$:

*the first one was waiting in the wings. this was poly-alphabeticsubstitution, in the form of the straight-alphabet vigenere with shortrepeating Leyword. the old **obEctions** to its use, which boiled down tothe impossibility of correcting a garbled dispatch **BuicLly** enough,vanished with the telegraph. it fulfilled the **reBuirements** of noncom-promisability of the general system and of ease of Ley changes. moreover,it had the reputation of being unbreaLable—which, if its cryptogramswere not divided into words, it largely was.*

Step 9. The last five letters each appear only a few times, and every word that contains them has only one possible reading: “BuicLly” and “reBuirements” are “quickly” and “requirements” ($B \rightarrow q$, $L \rightarrow k$); “Eefferson” and “obEctions” are “Jefferson” and “objections” ($E \rightarrow j$); “eYcept” is “except” ($Y \rightarrow x$); and “specialiMed” is “specialized” ($M \rightarrow z$). Now every letter of the cryptogram is decrypted:

the first one was waiting in the wings. this was poly-alphabeticsubstitution, in the form of the straight-alphabet vigenere with shortrepeating keyword. the old objections to its use, which boiled down tothe impossibility of correcting a garbled dispatch quickly enough,vanished with the telegraph. it fulfilled the requirements of noncom-promisability of the general system and of ease of key changes. moreover,it had the reputation of being unbreakable—which, if its cryptogramswere not divided into words, it largely was.

3.4 Program Output

Figures 3–5 show the program running in the Windows console. Figure 4 also shows that a wrong guess is rejected: after $V \rightarrow e$ was fixed, the guess $B \rightarrow e$ is refused, because two ciphertext letters cannot stand for the same plaintext letter.

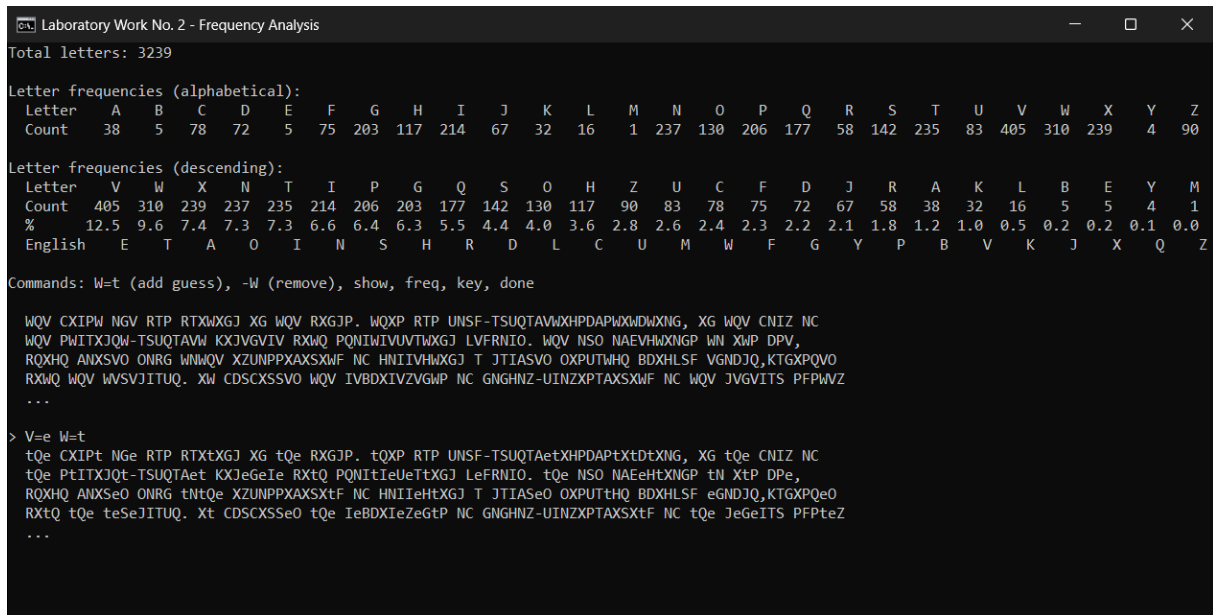


Figure 3: Frequency tables of the cryptogram and the first substitution.

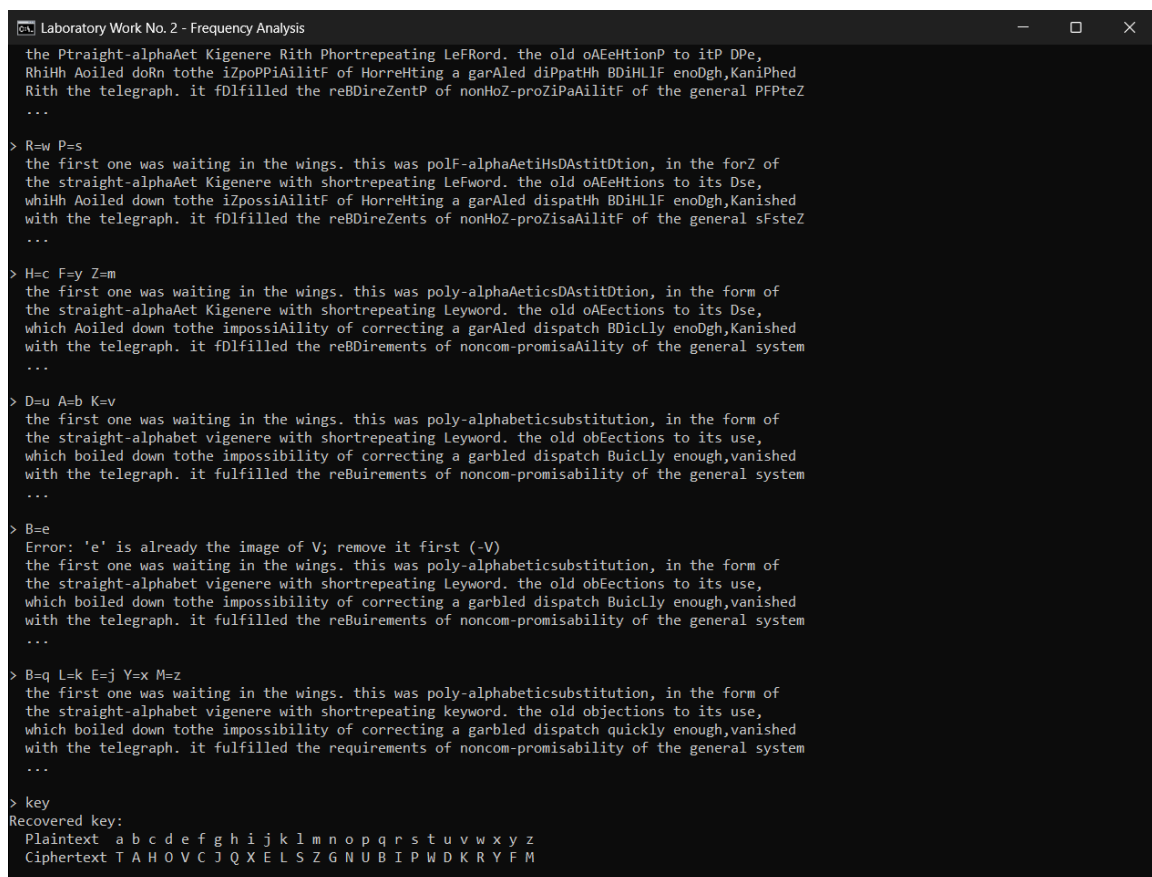


Figure 4: The last substitution steps, a rejected guess and the recovered key.

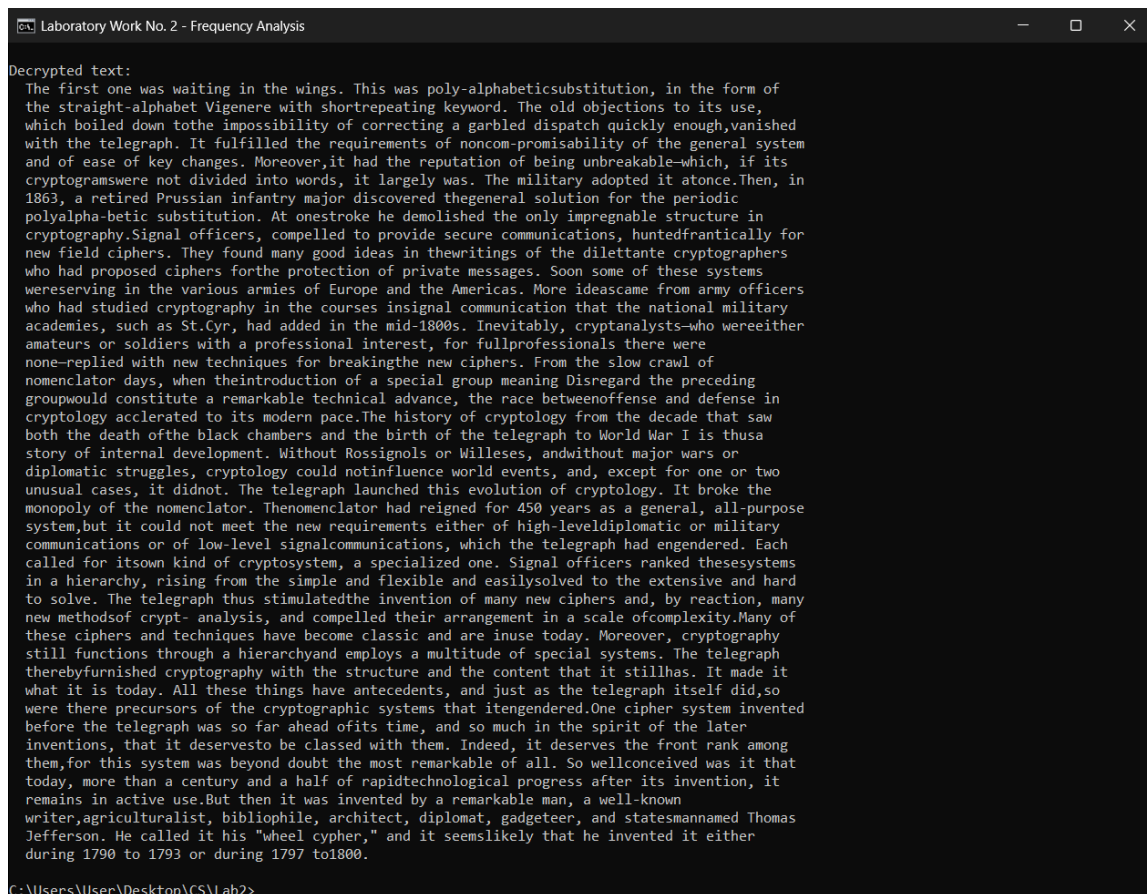


Figure 5: The decrypted message.

3.5 Results

The decrypted message (with the original upper and lower case restored) is:

The first one was waiting in the wings. This was poly-alphabetic substitution, in the form of the straight-alphabet Vigenere with short repeating keyword. The old objections to its use, which boiled down to the impossibility of correcting a garbled dispatch quickly enough, vanished with the telegraph. It fulfilled the requirements of noncom-promisability of the general system and of ease of key changes. Moreover, it had the reputation of being unbreakable—which, if its cryptograms were not divided into words, it largely was. The military adopted it at once. Then, in 1863, a retired Prussian infantry major discovered the general solution for the periodic polyalpha-betic substitution. At one stroke he demolished the only impregnable structure in cryptography. Signal officers, compelled to provide secure communications, hunted frantically for new field ciphers. They found many good ideas in the writings of the dilettante cryptographers who had proposed ciphers for the protection of private messages. Soon some of these systems were serving in the various armies of Europe and the Americas. More ideas came from army officers who had studied cryptography in the courses in signal communication that the national military academies, such as St. Cyr, had added in the mid-1800s. Inevitably, cryptanalysts—who were either amateurs or soldiers with a professional interest, for full professionals there were none—replied with new techniques for breaking the new ciphers. From the slow crawl of nomenclator days, when the introduction of a special group meaning Disregard the preceding group would

constitute a remarkable technical advance, the race between offense and defense in cryptology accelerated to its modern pace. The history of cryptology from the decade that saw both the death of the black chambers and the birth of the telegraph to World War I is thus a story of internal development. Without Rossignols or Willesees, and without major wars or diplomatic struggles, cryptology could not influence world events, and, except for one or two unusual cases, it did not. The telegraph launched this evolution of cryptology. It broke the monopoly of the nomenclator. The nomenclator had reigned for 450 years as a general, all-purpose system, but it could not meet the new requirements either of high-level diplomatic or military communications or of low-level signal communications, which the telegraph had engendered. Each called for its own kind of cryptosystem, a specialized one. Signal officers ranked these systems in a hierarchy, rising from the simple and flexible and easily solved to the extensive and hard to solve. The telegraph thus stimulated the invention of many new ciphers and, by reaction, many new methods of cryptanalysis, and compelled their arrangement in a scale of complexity. Many of these ciphers and techniques have become classic and are in use today. Moreover, cryptography still functions through a hierarchy and employs a multitude of special systems. The telegraph thereby furnished cryptography with the structure and the content that it still has. It made it what it is today. All these things have antecedents, and just as the telegraph itself did, so were there precursors of the cryptographic systems that it engendered. One cipher system invented before the telegraph was so far ahead of its time, and so much in the spirit of the later inventions, that it deserves to be classed with them. Indeed, it deserves the front rank among them, for this system was beyond doubt the most remarkable of all. So well conceived was it that today, more than a century and a half of rapid technological progress after its invention, it remains in active use. But then it was invented by a remarkable man, a well-known writer, agriculturalist, bibliophile, architect, diplomat, gadgeteer, and statesman named Thomas Jefferson. He called it his “wheel cypher,” and it seems likely that he invented it either during 1790 to 1793 or during 1797 to 1800.

The text is an excerpt about the influence of the telegraph on cryptography and about Thomas Jefferson’s “wheel cypher”; it appears to come from David Kahn’s book *The Codebreakers*. Words that are joined together, such as “to the” or “cryptograms were”, were already joined in the ciphertext.

Placing the plaintext alphabet next to the ciphertext alphabet gives the key that was used for encryption (Table 4). It is useful if other messages from the same sender are intercepted, since they are likely to use the same key.

Table 4: Reconstructed alphabet of the encrypted message (the key).

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	T	A	H	O	V	C	J	Q	X	E	L	S	Z	G	N	U	B	I	P	W	D	K	R	Y	F	M

4 CONCLUSION

Within this laboratory work, a monoalphabetic substitution cipher was broken by a frequency analysis attack, without any knowledge of the key. The intercepted cryptogram of variant 16 (3239 letters) was fully decrypted in nine steps, and the whole key, a permutation of the 26 English letters, was recovered.

Letter frequencies give the first foothold. The two most frequent ciphertext letters, V (12.5%) and W (9.6%), matched E (12.70%) and T (9.06%) exactly. For the following letters the frequencies alone were not enough: X, N, T, I, P and G all have frequencies between 6.3% and 7.4%, so their order did not match the English order A, O, I, N, S, H. Frequencies only narrowed the choice; the final decision came from the words.

The structure of the language does the rest. The most useful clues were the frequent short words (“the”, “and”, “of”, “a”, “it”, “in”) and long words with a unique pattern, such as “telegraph” and “cryptography”, which gave four letters at once. After about half of the letters were known, the remaining ones followed almost automatically from the context. The rarest letters (J, Q, X, Z) could only be found this way, because they appear just a few times.

Monoalphabetic ciphers are insecure regardless of the size of their key space. Although there are $26! \approx 4 \cdot 10^{26}$ possible keys, a text of a few thousand letters was broken in a few steps, because the cipher preserves the statistics of the language. The attack would be harder for a very short message, for a text without spaces between the words, or for a polyalphabetic cipher such as the Vigenère cipher, which is the subject of the decrypted text itself.

In conclusion, during this work I learned how to perform a frequency analysis attack, how to combine statistics with the knowledge of the language, and how to automate the routine part of the work with a small program, while the guesses are still made by the analyst.

5 SOURCE CODE

The full source code is available on GitHub: https://github.com/veacheslavv/CS_Labs/tree/main/Lab2